Architecting the Oideachais Knowledge Graph: Rapid PDF Indexing and Semantic Extraction Utilizing Z.ai MCP Servers within a 48-Hour Constraint

The intersection of generative artificial intelligence and advanced data engineering provides unprecedented opportunities to decode complex, unstructured educational corpora. Within the context of the kings_college_galway repository, the oideachais (education) data stream represents a highly sophisticated platform that merges multiple data pipelines into a single observable system.[1] This infrastructure, originally conceptualized as a Gemini Hackathon submission for analyzing GCSE, A-Levels, Leaving Cycle, and Junior Cycle materials, has evolved into a robust production environment.[1] It currently manages over thirty-seven distinct data assets via Dagster orchestration, covering cross-domain datasets encompassing Ireland, the United Kingdom, and geospatial data.[1] A critical and immediate imperative for this platform is the automated ingestion, semantic indexing, and structured understanding of the vast repositories of pedagogical documents hosted by the State Examinations Commission (examinations.ie) and the National Council for Curriculum and Assessment (ncca.ie). These documents, which include curriculum syllabi, past examination papers, and detailed marking schemes, hold the key to understanding the deep pedagogical patterns and assessment metrics required to prepare analytical models for the upcoming 2026 examination season.[2]
The execution of this initiative faces a stringent operational constraint: the maximization of a Z.ai Pro Coding Plan subscription, which is scheduled to expire in exactly forty-eight hours.[5]
This Pro tier subscription provides access to the frontier-class GLM-5.1 and GLM-5-Turbo foundation models, alongside highly specialized Model Context Protocol (MCP) servers, including the Vision MCP, Web Reader MCP, Web Search MCP, and Zread MCP.[5] Executing a pipeline of this magnitude within such a narrow timeframe necessitates a highly parallelized, resource-optimized sprint. The system architecture must perfectly synchronize the existing sruth/oideachais Dagster pipelines, the CocoIndex incremental processing framework, and the high-throughput capabilities of the GLM-5.1 model.[1] The ultimate objective is to transform decades of static, unstructured PDF archives into an active, queryable semantic knowledge graph that maps exact examination questions to their corresponding syllabus learning outcomes and grading rubrics.

Analyzing the Existing Repository Infrastructure: sruth/oideachais and docs/data_engineering

To successfully integrate the new ingestion and indexing logic, the architecture must maintain absolute fidelity to the structural and philosophical design patterns already established within the kings_college_galway repository. The project operates on a declarative, asset-oriented programming model orchestrated by Dagster, augmented by the Data Load Tool (DLT) and CocoIndex for data transformation and semantic indexing.[1]

The repository's internal directory structure heavily dictates the deployment strategy. The docs/data_engineering directory serves as the epistemological core of the project, housing the architectural decision records, data models, and extraction schemas that define how educational metadata is governed. Any new ontological frameworks designed to map the Irish educational system—such as the relational schemas linking a State Examinations Commission (SEC) marking scheme to a National Council for Curriculum and Assessment (NCCA) learning outcome—must be rigorously documented within this directory. This ensures that the data engineering principles remain transparent and reproducible for future contributors or autonomous agents operating within the repository.

The executable logic resides within the sruth/oideachais directory. In the Irish language, "sruth" translates to "stream," indicating that this directory houses the active data ingestion and transformation pipelines specific to the educational domain. The existing codebase relies on Python-based asynchronous processing and relies on the execution command cd education dagster dev to initiate the local development server and materialize the thirty-seven existing assets.[1] The immediate engineering task involves expanding this directory to include new Dagster Software-Defined Assets (SDAs) specifically dedicated to navigating the SEC and NCCA portals, downloading the nested PDF files, utilizing visual parsing libraries to convert these documents into Markdown, and invoking the Z.ai APIs to execute semantic extraction. Because the repository already utilizes CocoIndex for incremental processing and semantic search capabilities [1], the new pipeline components will slot seamlessly into the existing graph, ensuring that only net-new or modified examination papers are processed, thereby conserving precious API quotas.

Directory Path	**Functional Purpose within the Repository**	**Expansion Requirements for the 48-Hour Sprint**
docs/data_engineering	Architectural documentation, schema definitions, and data governance policies.	Addition of relational schemas defining the hierarchy between NCCA Learning Outcomes and SEC Examination Rubrics.
sruth/oideachais	Executable Python pipelines, Dagster asset definitions, and CocoIndex flow configurations.	Implementation of asynchronous PDF scrapers, Vision MCP OCR parsers, and GLM-5.1 semantic extraction assets.
bunchloch	Foundational data, including University of Galway notes and raw reference materials.[8]	Storage protocols for the raw PDF binaries retrieved from the state examination portals prior to processing.

Exploiting the Z.ai Pro Coding Plan and GLM-5.1 Architecture

The engine driving the semantic extraction phase of this project is the Z.ai Pro Coding Plan, anchored by the GLM-5.1 foundation model. Understanding the exact technical specifications, capabilities, and constraints of this subscription tier is the single most critical factor in guaranteeing the success of the forty-eight-hour engineering sprint.

The GLM-5.1 model is a flagship foundation model optimized for complex systems engineering and long-horizon agentic tasks.[9] It boasts an immense context length of 200,000 tokens and a maximum output generation capacity of 128,000 tokens.[10] This massive context window is an absolute necessity for processing educational corpora. A single Leaving Certificate examination paper, combined with its corresponding marking scheme and the overarching subject syllabus, can easily exceed tens of thousands of tokens when converted to text. The 200K window allows the model to hold all three documents in its working memory simultaneously, performing deep cross-referencing to determine exactly how a specific grading rubric aligns with a theoretical learning outcome without suffering from context degradation or retrieval-induced hallucinations.[11] Empirical testing demonstrates that GLM-5.1 achieves state-of-the-art performance on benchmarks such as SWE-Bench Pro and exhibits highly reliable performance for long-range tasks across extensive context windows.[11]

However, the raw power of the model is heavily moderated by the strict usage quotas imposed by the Pro Coding Plan. The infrastructure is designed for high-frequency development but implements rolling limits to manage server load. The Pro Plan provides a dynamic limit of approximately 400 prompts every five hours, alongside a hard weekly cap of 2,000 prompts.[5] It is critical to note that a single "prompt" in the context of an agentic workflow often translates to 15 to 20 actual model invocations, as the agent thinks, plans, executes tools, and verifies its output.[5]

Furthermore, the consumption of these prompt quotas is subject to time-based multipliers. The GLM-5.1 and GLM-5-Turbo models are highly resource-intensive. During peak hours, defined as 14:00 to 18:00 UTC+8, usage is deducted at a multiplier of three times the standard rate.[5] During off-peak hours, the deduction drops to a two-times multiplier, with periodic promotional periods reducing off-peak consumption to a one-to-one ratio.[5] Given the 48-hour deadline, the data engineering team must architect the Dagster execution schedules to align strictly with these off-peak windows. Mass batch processing of the PDFs—where the pipeline concatenates multiple documents into single, massive JSON extraction requests to maximize the 200K context window—must be cron-scheduled to avoid the 14:00 to 18:00 UTC+8 peak window. This strategic orchestration effectively triples the volume of data that can be processed before the subscription expires.

The Model Context Protocol (MCP) Paradigm

Beyond the foundation model itself, the Z.ai Pro Coding Plan unlocks an ecosystem of remote and local Model Context Protocol (MCP) servers. The MCP is an open standard that provides a universal translation layer, allowing Large Language Models to interact securely and seamlessly with external data sources, local file systems, and specialized APIs.[15]

The protocol operates by separating the standardized JSON-RPC 2.0 message structure from the underlying transport mechanisms.[17] Whether the client (such as Claude Code, Cursor, or a custom Python script) is communicating with a local subprocess via standard input/output (stdio) or communicating with a remote server via HTTP Server-Sent Events (SSE), the protocol semantics remain identical.[17] The MCP architecture defines three primary capabilities: Resources (file-like data that the model can read), Prompts (pre-written templates to guide the model), and Tools (executable functions the model can invoke).[16]

The Z.ai Pro Plan provides access to four highly specialized MCP servers that will be instrumental in constructing the oideachais pipeline. The Zread MCP Server is designed for deep repository comprehension.[18] Powered by zread.ai, it connects via HTTP and provides tools such as search_doc, get_repo_structure, and read_file.[18] The Web Reader MCP Server allows the model to fetch and extract full-page content and structured metadata from specified URLs.[19] The Web Search MCP Server provides real-time information retrieval capabilities.[20] Finally, the Vision MCP Server provides powerful optical character recognition and image analysis capabilities using the GLM-4.6V visual language model.[21]

These MCP servers are subject to their own distinct quota restrictions. The Pro Plan allocates a combined total of 1,000 web searches, web reads, and Zread calls per month.[5] The Vision MCP, however, draws directly from the 5-hour dynamic prompt resource pool associated with the core LLM package.[5] This quota separation mandates a highly tactical deployment strategy. The Web Reader and Zread MCPs must be used sparingly, reserved exclusively for the initial phase where the AI agent is writing the pipeline code and exploring the repository. Once the pipeline enters the mass data extraction phase, it must rely on standard Python libraries for web requests and reserve the Vision MCP solely for processing visually complex PDFs that standard parsers cannot handle.

Z.ai MCP Server	Core Capabilities & Tools	Transport Protocol	Pro Plan Quota Constraints	Role in the 48-Hour Pipeline Sprint
Zread MCP	get_repo_structure, read_file, search_doc	Streamable-HTTP / SSE	Shared pool: 1,000 calls/month	Analyzing the existing sruth/oideachais Dagster assets to ensure new code matches existing stylistic paradigms.[18]
Web Reader MCP	webReader	Streamable-HTTP / SSE	Shared pool: 1,000 calls/month	Dynamically analyzing the

				DOM structure of examinations.ie to generate URL routing logic for Python scrapers.[19]
Web Search MCP	webSearchPrime	Streamable-HTTP / SSE	Shared pool: 1,000 calls/month	Resolving unknown variables or finding current NCCA policy updates regarding the Senior Cycle redevelopment.[20]
Vision MCP	extract_text_from_screenshot, ui_to_artifact, understand_technical_diagram	Stdio (Local execution)	Shares the 400 prompt / 5-hour rolling pool	Performing high-fidelity OCR and structural extraction on complex PDF layouts, such as physics diagrams or nested mathematics.[21]

Agentic Orchestration and Rapid Pipeline Development

The first twelve hours of the 48-hour sprint must be entirely dedicated to utilizing the AI agent to write the ingestion and extraction scripts. The developer will interface with a CLI-based agentic coding tool, such as Claude Code, Cline, or OpenCode, which has been authenticated against the Z.ai endpoints and configured to utilize the GLM-5.1 model.[22]

The agent is first directed to invoke the Zread MCP server to ingest the architectural context of the kings_college_galway repository. By executing get_repo_structure and reading the files within docs/data_engineering and sruth/oideachais, the agent internalizes the project's reliance on Dagster for orchestration, DLT for loading, and CocoIndex for semantic processing.[1] This ensures that when the agent generates new Python scripts for scraping the state examination portals, the code will seamlessly integrate into the existing framework, utilizing the correct environment variables, logging standards, and asynchronous programming paradigms.

Simultaneously, the agent will utilize the Web Reader MCP to navigate the intricate and highly specific directory structures of the target websites. The State Examinations Commission maintains an Examination Material Archive that houses past papers and marking schemes dating back to 1993.[23] The archival URLs follow highly deterministic patterns based on the examination year, the examination type (e.g., Junior Certificate or Leaving Certificate), the subject, and the level (Higher, Ordinary, Foundation, or Common).[25] However, accessing these archives programmatically requires bypassing specific terms-and-conditions acceptance gates.[24] The agent, utilizing the Web Reader, will analyze the required HTTP POST payloads and generate a robust, asynchronous Python scraper using libraries such as httpx or aiohttp.

The National Council for Curriculum and Assessment (ncca.ie) presents a different structural challenge. Curriculum specifications and syllabi are hosted across various subdomains and often categorized by the educational phase (Early Childhood, Primary, Junior Cycle, Senior Cycle).[28] The agent will utilize the Web Reader to locate the specific download links for the finalized PDF syllabi, filtering out draft documents or consultation reports.[29]

Crucially, once the agent has generated these robust Python scraping scripts, the pipeline completely ceases its reliance on the Web Reader MCP. By transitioning the mass-downloading phase to standard Python asynchronous execution, the project entirely circumvents the 1,000-call monthly quota limit imposed on the MCP servers, preserving these resources for absolute emergencies or complex debugging tasks later in the sprint. The output of this initial phase is a local directory, highly organized by year, subject, and document type, containing the thousands of raw binary PDF files ready for the data engineering pipeline.

Ontological Topography: Deciphering the NCCA Syllabi

Before the downloaded PDFs can be semantically extracted, the system requires a rigid ontological schema that defines what data is valuable and how it relates to other entities within the knowledge graph. This requires a profound understanding of the pedagogical philosophies underpinning the Irish educational system.

The documents produced by the National Council for Curriculum and Assessment (ncca.ie) define the theoretical expectations of the student. Historically, the Irish educational system relied on highly technical, content-heavy syllabi that dictated exactly what a teacher should cover in the classroom.[32] However, modern educational reform, particularly the redevelopment of the Junior Cycle (implemented from 2015 onwards) and the ongoing redevelopment of the Senior Cycle (2025-2029), has drastically shifted this paradigm.[2]

Modern NCCA curriculum specifications are no longer driven by rigid content lists, but by Learning Outcomes.[37] A Learning Outcome is an explicit statement that describes the understanding, skills, and values a student should be able to demonstrate after a period of learning.[37] These outcomes are deeply hierarchical. At the highest level, the curriculum is governed by overarching Key Competencies (e.g., "Thinking and Learning," "Communicating," "Being Creative").[38] These competencies map down into specific subject Strands, which further subdivide into Elements, and finally terminate at the specific Learning Outcome Labels and individual Learning Outcomes.[39]

The phrasing of a Learning Outcome is highly deliberate and relies on specific pedagogical action verbs. Verbs such as "Investigate," "Assess," "Compare," or "Critique" dictate the cognitive depth and the type of assessment required.[28] For example, a Learning Outcome asking a student to "Critique" an idea requires balancing positive and negative aspects, whereas an outcome asking them to "Compare" requires an account of similarities and differences.[28]

The data engineering challenge involves instructing the LLM to parse the syllabus PDFs and reconstruct this exact hierarchy in a structured JSON format. The schema must enforce relational integrity, ensuring that a specific Learning Outcome (e.g., "Investigate the relationships between image and sound") is correctly mapped to its parent Element, Strand, and Key Competency.[28] Furthermore, because the NCCA is currently in a transitional phase—with older Leaving Certificate subjects utilizing traditional content-based syllabi while newer specifications utilize the Learning Outcome framework—the extraction prompt must be sufficiently dynamic to recognize the document's era and apply the appropriate ontological template.[33]

Structural Analysis of State Examinations and Marking Schemes

While the NCCA documents dictate the theory of education, the archives of the State Examinations Commission (examinations.ie) dictate the reality of assessment. The SEC is responsible for generating the examination papers, overseeing the marking process, and publishing the finalized marking schemes and Chief Examiners' reports.[42] The parsing of these documents presents the most significant technical hurdle of the entire project due to their highly non-standardized and complex visual structures.

Examination papers are notoriously difficult for standard Optical Character Recognition (OCR) engines to process. A Higher Level Mathematics or Physics paper is replete with complex algebraic matrices, calculus notation, and technical diagrams.[32] A Geography paper may feature intricate topographical maps, while an Art paper may include high-resolution historical paintings.[46] Furthermore, many examinations, particularly in the humanities, are published bilingually, presenting the Irish (Gaeilge) text adjacent to the English translation in complex multi-column layouts.[48] A naive PDF-to-text parser will inevitably scramble these columns, separating mathematical equations from their corresponding question identifiers and destroying the semantic integrity of the document.

The marking schemes published by the SEC are even more structurally volatile. As explicitly stated in the SEC's internal documentation, marking schemes are "working documents" that are not finalized until examiners have applied a draft scheme to a sample of candidates' work and collated feedback.[42] They are not intended to be standalone guides, but rather essential resources for trained examiners.[42] Consequently, their layout is highly idiosyncratic to the subject and the specific Chief Examiner of that year.

For instance, in Leaving Certificate History, the marking scheme mandates a strict division of points. A typical essay question is evaluated under two distinct headings: the Cumulative Mark (CM), which awards points for accurate and relevant historical content, and the Overall Evaluation (OE), which assesses the quality, structure, and coherence of the response.[49] The proportion of CM to OE generally remains at a 60/40 ratio.[49] In other subjects, the marking scheme may utilize specialized "A-scales" that dictate specific point deductions for incorrect responses or partial credit scenarios.[51] In more subjective subjects, such as Agricultural Science or the Physical Activity Project in Physical Education, the marking scheme will outline extensive qualitative rubrics for evaluating a student's Individual Investigative Study or video submission.[52]

To successfully map the knowledge graph, the pipeline must extract not just the text of the marking scheme, but the mathematical logic of its grading rubric. The GLM-5.1 model must be instructed to parse the non-standard grids, isolate the point allocations, define the acceptable alternative answers, and quantify the penalty structures, converting this highly subjective examiner notation into a standardized, machine-readable JSON object that accurately reflects the difficulty and stringency of that specific examination year.

Integrating Dagster Orchestration and CocoIndex Incremental Processing

To manage the transformation of thousands of complex PDFs into structured JSON within the 48-hour constraint, the architecture heavily relies on the synergy between Dagster and CocoIndex.[1] This integration ensures that the data pipeline is not only observable and robust but also fundamentally optimized to minimize unnecessary API consumption.

CocoIndex is a high-performance data processing framework explicitly designed for AI-native applications.[54] Its defining feature is incremental processing. When building semantic pipelines, re-processing an entire corpus of PDFs through an LLM every time a single file is updated is prohibitively expensive and time-consuming. CocoIndex solves this by tracking the state of the source data. By defining a cocoindex.FlowBuilder and pointing it to the local directory containing the downloaded SEC and NCCA PDFs, the framework ensures that only net-new documents or files that have been explicitly modified are pushed through the transformation steps.[54] If an older syllabus is removed from the directory, CocoIndex automatically purges the corresponding Markdown and JSON extractions from the downstream vector store.[54]

The integration of CocoIndex within the Dagster environment represents a highly advanced data engineering pattern.[56] Each stage of the ETL process is defined as a distinct Dagster Software-Defined Asset (SDA). The pipeline follows a rigid execution graph:

1. **Ingestion:** The asynchronous Python scrapers execute, updating the local raw PDF directory.
2. **Conversion:** The CocoIndex flow detects the new PDFs and initiates the conversion to Markdown.
3. **Extraction:** The Markdown representations are batched and submitted to the GLM-5.1 API for structured JSON extraction.
4. **Ontological Linking:** The extracted exam questions are semantically mapped against the extracted syllabus learning outcomes.
5. **Materialization:** The final relational graph is loaded into the target database or vector store.

The conversion stage is the most critical bottleneck. To prevent the destruction of complex examination layouts, the pipeline utilizes intelligent routing. Standard, text-heavy

documents—such as Chief Examiners' reports or simple language syllabi—are processed locally using high-speed open-source parsers like Docling or Marker, which can be wrapped as custom FunctionSpec classes within the CocoIndex flow.[57]
However, when the pipeline encounters a document with a high visual density score—such as a Higher Level Mathematics paper or a complex Engineering drawing—it dynamically routes the PDF page to the local Z.ai Vision MCP Server.[21] Powered by the GLM-4.6V vision model, the MCP server utilizes tools like extract_text_from_screenshot and understand_technical_diagram to perform high-fidelity OCR, preserving the spatial relationships of equations, deciphering architectural flowcharts, and translating complex tables into perfectly formatted Markdown grids.[21] The GLM-OCR capability is exceptionally efficient, achieving a throughput of 1.86 pages per second for PDF documents, ensuring that even complex visual parsing does not disrupt the aggressive 48-hour timeline.[62]
Furthermore, the entire Dagster environment can be exposed directly back to the orchestrating AI agent via the mcp-server-dagster implementation.[63] This creates a powerful bidirectional feedback loop. The AI agent, operating within Claude Code or Cline, can query the Dagster GraphQL endpoint to list available assets, monitor recent run logs, and manually trigger job executions.[66] If a specific batch of marking schemes fails the CocoIndex conversion step due to an unforeseen layout anomaly, the agent receives the error log, autonomously refactors the Python parsing script, and triggers a re-materialization of the failed asset, all without requiring human intervention.[66]

Semantic Extraction Strategies and Token Optimization

With the educational corpora successfully converted into pristine Markdown, the pipeline enters its most computationally intensive phase: semantic extraction via the GLM-5.1 foundation model. The objective is to transform the linear text of the syllabi and examination papers into a highly relational, queryable knowledge graph.
To enforce absolute schema conformity and eliminate LLM hallucination, the CocoIndex extraction flow is tightly coupled with a typed prompt engineering framework, such as BAML (BoundaryML) or Pydantic.[68] These frameworks allow the data engineer to define strict data models that the GLM-5.1 model is forced to populate.[68] By transforming prompts into strongly typed functions, the system guarantees that the resulting JSON payload matches the expected schema, preventing downstream database insertion failures.
The extraction process operates in two distinct phases. Phase one is isolation and categorization. The LLM processes an examination paper and fragments it into individual, atomic questions. It extracts the metadata (Year, Subject, Level), the exact text of the question, any associated diagrams (referenced via Markdown image tags generated by the Vision MCP), and the total marks allocated. Simultaneously, it processes the corresponding marking scheme, isolating the specific grading rubric, the A-scale penalties, and the acceptable alternative answers for that exact question.[42]
Phase two is the semantic synthesis. The LLM is provided with the isolated examination

question, its grading rubric, and the complete list of NCCA Learning Outcomes for that specific subject. The model is instructed to perform deep semantic reasoning to link the question to the precise pedagogical intention it was designed to assess.[28] This requires the model to understand the nuances of the pedagogical verbs; it must recognize that an exam question asking a student to "Evaluate the historical development of policy reforms" [70] directly maps to a Learning Outcome requiring students to "Critique" or "Assess" historical data, rather than simply "Identify" it.[28]

Executing this synthesis at scale requires meticulous token management to avoid exhausting the 400 prompt / 5-hour rolling limit of the Pro Coding Plan.[5] The pipeline leverages the massive 200,000-token context window of the GLM-5.1 model to process data in massive batches.[10] Rather than sending one question per prompt, the CocoIndex flow concatenates an entire examination paper, its marking scheme, and the relevant syllabus section into a single, highly structured payload. The model processes the entire document matrix in one inference pass, returning a massive JSON array of perfectly linked entities. This batching strategy effectively processes hundreds of pages of educational data while consuming only a single prompt from the quota.[5]

Furthermore, the execution schedule is heavily optimized around the Z.ai infrastructure multipliers. Because the GLM-5.1 model consumes quota at a rate of 3x during peak hours (14:00–18:00 UTC+8) and 2x (or 1x during promotional periods) during off-peak hours, the Dagster orchestrator is configured to pause all semantic extraction tasks during the peak window.[5] During these four hours, the pipeline focuses exclusively on the local CocoIndex Markdown conversion and PDF downloading, hoarding the extracted text. Once the off-peak window resumes, the pipeline aggressively flushes the queue to the GLM-5.1 API, effectively doubling or tripling the total volume of data processed within the 48-hour timeframe.[11]

Overcoming Edge Cases: Bilingual Contexts and Dynamic Assessment Models

Operating a data pipeline of this complexity at extreme velocity guarantees the emergence of significant edge cases, primarily driven by the unique linguistic requirements and evolving assessment models of the Irish educational system.

The most pervasive challenge is the bilingual nature of the state examinations. A significant portion of the Irish student body undertakes their examinations through the medium of Irish (Gaeilge) across all subjects, not just language courses.[30] Consequently, many syllabi and examination papers present the content in both Irish and English simultaneously.[48] If the foundation model lacks robust multilingual proficiency, it will inevitably corrupt the translation or fail to recognize the semantic equivalence between the Irish and English phrasing of a specific learning outcome.

Fortunately, the GLM series of models—particularly GLM-5.1 and GLM-4.7—are engineered with exceptionally strong multilingual capabilities, ensuring solid performance across reasoning and knowledge benchmarks regardless of the language inputted.[9] The system prompt within the CocoIndex extraction flow is specifically calibrated to recognize bilingual documents. The

LLM is instructed to extract the semantic meaning independent of the language, tagging the output JSON to indicate whether the source text was in Irish, English, or both, thereby allowing the resulting knowledge graph to support native querying in either language.

A secondary edge case involves the physical scale and subjectivity of certain examination formats. The Irish educational system is undergoing a massive redevelopment, moving away from a singular reliance on terminal written examinations. The Senior Cycle reform (2025-2029) mandates that all subjects must include a 40% project assessment component.[2] Many subjects, such as Design and Communication Graphics, Art, and Agricultural Science, already rely heavily on extensive coursework, portfolios, and Individual Investigative Studies (IIS).[53] Furthermore, the introduction of the Physical Education Leaving Certificate subject incorporates the assessment of a Physical Activity Project via video submission.[52]

The marking schemes for these components are vastly different from standard written papers. They describe highly subjective, qualitative evaluation criteria regarding technical execution, aesthetic presentation, or physical performance.[52] The extraction schema must dynamically adapt when it encounters these subject types. The LLM utilizes its advanced reasoning capabilities to interpret these qualitative rubrics, extracting the core evaluative parameters and converting the subjective examiner notes into standardized metrics that can be queried and analyzed alongside traditional mathematical grading scales.[10]

Strategic Synthesis and Future Implementation

The successful execution of this architectural sprint within the 48-hour Z.ai Pro Coding Plan window establishes a permanent, continuously self-updating foundation for the kings_college_galway repository. By synthesizing the orchestration power of Dagster, the incremental processing efficiency of CocoIndex, and the frontier-class semantic extraction capabilities of the GLM-5.1 model, the project successfully transforms decades of static, bureaucratic PDFs into a dynamic, interrogable intelligence platform.

The integration of the Model Context Protocol (MCP) servers proves to be a paradigm-shifting methodology for data engineering. By delegating the codebase analysis to the Zread MCP, the web scraping logic to the Web Reader MCP, and the complex PDF OCR to the Vision MCP, the AI agent can autonomously construct the entire ingestion pipeline, leaving the developer to focus purely on the higher-order ontological schema design.[75] Furthermore, because CocoIndex tracks the state of all processed files, the system is completely future-proofed. When the 2026 examination season concludes and the State Examinations Commission publishes the new wave of examination papers and marking schemes, the Dagster pipeline will ingest and parse the new documents automatically, requiring only a fraction of the computational resources utilized during this initial historical backfill.[3]

The implications of this resulting knowledge graph for the oideachais platform are profound. It provides educators, policymakers, and students with unprecedented empirical visibility into the mechanics of the Irish educational system. By querying the semantic database, users can instantly determine which NCCA Learning Outcomes are statistically over-examined, how the phrasing of assessment criteria shifts longitudinally in response to Chief Examiners' reports, and how grade boundaries fluctuate across specific subjects.[43]

As the educational landscape transitions heavily toward continuous assessment and project-based learning under the Senior Cycle redevelopment [2], this historical analysis provides the crucial baseline required to navigate the new paradigms. Furthermore, the digitization of the complex SEC marking schemes into machine-readable JSON objects paves the way for the development of autonomous AI grading systems. Secondary agents can evaluate student practice submissions against the exact historical standards and rubrics employed by the State Examinations Commission, providing instantaneous, highly objective feedback, and ultimately fulfilling the original mandate of the kings_college_galway educational platform.

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